

Many-body perturbation theory for the nuclear EoS up to fifth order

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- Summary

Limitations of MBPT in Nuclear Matter:

- Insufficient Order.
- Incomplete treatment of 3N forces.
- Unclear convergence.

What has this article done?

- An automated, GPU-accelerated framework for MBPT calculations.
- Analyze the convergence of MBPT in PNM and SNM. Study on the saturation properties of asymmetric nuclear matter.
- Lay the foundation for harder interactions and non-perturbative benchmarks.

Many-body expressions

Total energy density of nuclear matter can be expressed as

$$E_{\text{tot}} = E^{(\text{HF})} + E^{(2)} + E_1^{(3)} + E_2^{(3)} + E_3^{(3)} + \dots \quad (1)$$

where

- E^{HF} is the mean-field energy.
- $E^{(2)}$ is the second-order correlation energy.
- $E_{1,2,3}^{(3)}$ are the contributions from three third-order Goldstone diagrams with distinct topologies, respectively.

Consideration of three-nucleon forces:

$$\frac{E^{(\text{HF})}}{V} = \sum_i \epsilon_{\tau_i}(k_i) - \frac{1}{2} \sum_{ij} \langle ij | \tilde{V}_{\text{eff}}^{(\xi=2/3)} | ij \rangle \quad (2)$$

$$\epsilon_{\tau_i}(k_i) = \frac{k_i^2}{2m_n} + \sum_j \langle ij | \tilde{V}_{\text{eff}}^{(\xi=1/2)} | ij \rangle \quad (3)$$

$$\langle p_2 p_3 | \tilde{V}_{\text{eff}}^{(\xi)} | p_2 p_3 \rangle = \langle p_2 p_3 | \mathcal{A}_{12} V_{NN} | p_2 p_3 \rangle + \xi \sum_i \langle i p_2 p_3 | \mathcal{A}_{123} V_{3N} | i p_2 p_3 \rangle \quad (4)$$

level	MBPT(<i>n</i>)	$N_{\text{diag}}^{\text{a}}$	$N_{\text{diag}}^{(\text{eff})}$	N_{dim}	N_{grp}
two-body	HF	1	1	6	1
	2	1	1	9	1
	3	3	3	18	2
	4	39	24	24	4
	5	840	375	30	5
	6 ^b	27300	11269	36	6
three-body	2	1	1	18	1
	3	19	14 ^c	21; 24; 27	14

3N force cannot be calculated in terms of the effective two-body interaction (4) only, there are 15 terms of residual 3N force up to third order^a.

Figure: Number of Goldstone diagrams.

^aB. S. Hu, T. Li, and F. R. Xu, (2018), arXiv:1810.08804

Automated evaluation of many-body expressions

Notice before calculation:

- Beyond third order, MBPT expressions may appear in complex-conjugate pairs, and integrand functions may exhibit energy-denominator divergences that cancel order by order when added to the integrand function of the companion diagram.
- Higher and more controllable numerical accuracies can be obtained due to partial cancellations among the terms in each group, whose uncertainties would otherwise be assumed uncorrelated and summed in quadrature.
- Computationally demanding tasks, can be combined and thus more efficiently parallelized.

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	4	39	24	24	4
	5	840	375	30	5
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three-body	2	1	1	18	1
	3	19	14 ^c	21; 24; 27	14

Figure: Number of Goldstone diagrams.

Group MBPT diagrams order by order based on their particle-hole composition and then integrate the sums of their integrand functions. Monte Carlo integration is adopted, with the cutoff function being

$$f_{\text{NN}}(p) = \exp\left[-\frac{p^n}{\Lambda_{\text{NN}}^n}\right], \quad f_{3\text{N}}(p, q) = \exp\left[-\frac{(p^2 + 3q^2/4)^2}{\Lambda_{3\text{N}}^4}\right] \quad (5)$$

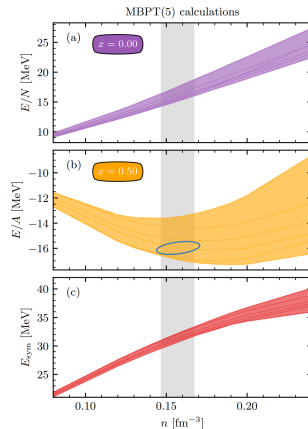
PNM and SNM calculations using MBPT(5)

Hebeler et al. interaction^a is considered, which includes two types of core interactions: the chiral low-momentum two-nucleon interaction and the improved chiral three-nucleon interaction. These interactions are softened and evolved through the renormalization group method.

The figure right shows:

- EoS of PNM is smooth.
- There is obvious saturation point of SNM.

^aK. Hebeler, S. K. Bogner, R. J. Furnstahl, A. Nogga, and A. Schwenk, Phys. Rev. C 83, 031301(R) (2011)

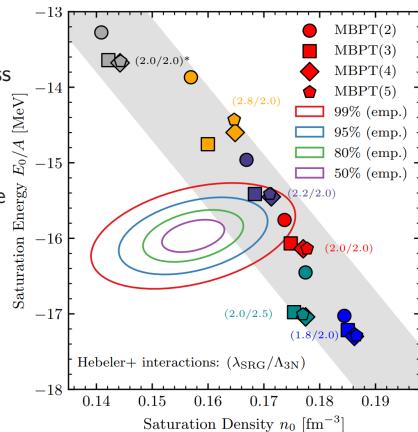


Nuclear saturation point

Extract all the SNM saturation points $(n_0, \frac{E_0}{A})$ obtained at each interaction and each MBPT level, and fit the curves using Gaussian Process to more reliably identify the minimum point.

Main results:

- The differences in $(n_0, \frac{E_0}{A})$ between successive MBPT orders decrease monotonically.
- All the results formed distinct Coester bands.
- Can not reproduce the empirical saturation density and energy.



Many-body convergence and residual 3N contributions up to third order

Hebeler *et al.* (2.0/2.0)

PD	HF	2nd	3rd	4th	5th	Total	Padé[2/2]	Padé[3/2]	Padé[2/3]
P08	9.88(00)	-0.70(01)	0.11(01)	-0.05(01)	0.01(01)	9.26(01)	9.26	9.26	9.26
P16	16.57(00)	-0.80(01)	0.10(01)	-0.06(01)	0.01(01)	15.82(02)	15.81	15.81	15.81
P30	29.39(00)	-0.85(01)	0.06(01)	-0.09(01)	0.03(02)	28.53(02)	28.51	28.52	28.52
S08	-8.09(00)	-3.75(01)	-0.36(01)	-0.01(01)	-0.07(01)	-12.28(02)	-12.20	-12.23	-12.29
S16	-12.02(00)	-3.61(01)	-0.30(01)	-0.05(01)	0.02(01)	-15.96(01)	-15.98	-15.97	-15.95
S30	-7.24(00)	-3.50(01)	-0.31(01)	-0.31(01)	-0.03(03)	-11.39(03)	-11.45	-11.42	-11.42

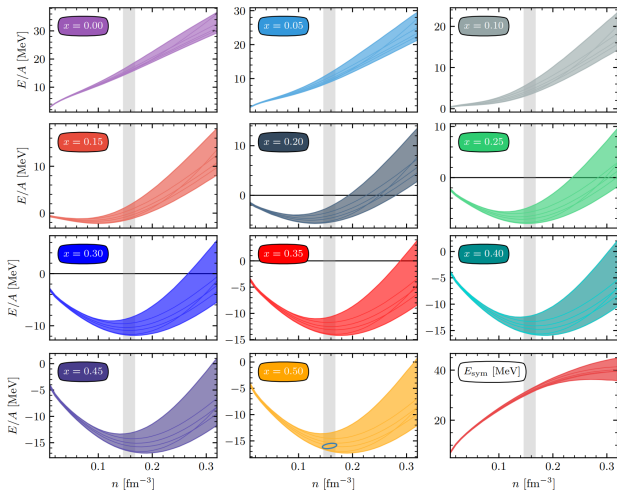
Hebeler *et al.* (2.8/2.0)

PD	HF	2nd	3rd	4th	5th	Total	Padé[2/2]	Padé[3/2]	Padé[2/3]
P08	10.77(00)	-1.43(01)	0.28(01)	-0.11(01)	0.03(01)	9.54(01)	9.53	9.53	9.53
P16	18.68(00)	-2.30(01)	0.38(01)	-0.18(01)	0.01(02)	16.60(02)	16.62	16.58	16.59
P30	34.32(00)	-3.62(01)	0.32(01)	-0.40(02)	0.05(02)	30.68(03)	30.65	30.63	30.63
S08	-4.55(00)	-6.55(01)	-0.71(01)	0.25(01)	-0.00(01)	-11.56(02)	-11.51	-11.57	-11.58
S16	-5.75(00)	-8.07(01)	-0.90(01)	0.16(01)	0.15(02)	-14.40(02)	-14.51	-14.31	-14.38
S30	2.88(00)	-9.77(01)	-1.00(01)	-0.66(02)	0.10(03)	-8.46(04)	-8.75	-8.53	-8.44

Figure: “P” (“S”) stands for PNM(SNM), and the following number D corresponds to the density $n = D/100\text{fm}^{-3}$

Explicit Asymmetric Nuclear Matter Calculations using MBPT(4)

- The EOS of neutron-rich matter is tightly constrained at $n \lesssim n_0$.
- Uncertainty increases simultaneously with both the density and the proton fraction.
- When $x \gtrsim 0.15$, the system gradually exhibits saturation properties.



DHS interactions using MBPT(4)

DHS interaction¹: Based on the EFT of the N³LO chirality interactions, including the chirality interactions of NN, 3N and 4N. Compared to the previous calculation, 4th normal-ordered 3N and 3rd residual 3N have been taken into consideration.

N³LO DHS (EMN 500 MeV), with $c_D = -3.00$ and $c_E = -2.22$

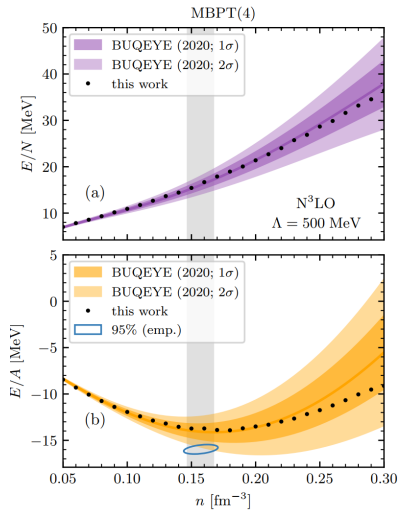
PD	HF	2nd (2b)	2nd (3b)	3rd (2b)	3rd (3b)	4th	Total	Padé[2/2]
P08	12.64(00)	-3.92(01)	-0.05(01)	0.57(01)	0.19(03)	-0.11(03)	9.33(04)	9.34
P16	23.67(00)	-7.95(01)	-0.08(01)	0.82(01)	0.26(05)	-0.03(05)	16.68(08)	16.67
P30	50.76(00)	-13.14(02)	-0.04(01)	0.50(04)	-0.10(09)	-1.55(09)	36.42(13)	36.32
S08	-1.42(00)	-8.97(01)	-0.32(01)	-0.36(01)	-0.40(01)	0.72(04)	-10.75(04)	-10.69
S16	1.71(00)	-14.63(01)	-0.84(01)	-0.99(02)	-0.08(04)	1.08(06)	-13.76(08)	-13.69
S30	21.48(00)	-25.43(02)	-1.38(01)	-0.83(04)	0.22(10)	-3.16(08)	-9.11(13)	-9.71

N²LO DHS (EMN 500 MeV), with $c_D = -1.75$ and $c_E = -0.64$

PD	HF	2nd (2b)	2nd (3b)	3rd (2b)	3rd (3b)	4th	Total	Padé[2/2]
P08	11.35(00)	-2.04(01)	-0.03(01)	0.38(01)	0.06(01)	-0.15(01)	9.56(01)	9.59
P16	21.14(00)	-4.07(01)	-0.04(01)	0.51(01)	0.09(02)	-0.30(02)	17.32(03)	17.37
P30	45.81(00)	-6.38(01)	-0.02(01)	0.93(02)	-0.14(03)	-0.93(04)	39.28(05)	39.38
S08	-1.82(00)	-8.74(01)	-0.30(01)	0.22(01)	-1.18(01)	0.63(01)	-11.20(02)	-11.10
S16	1.21(00)	-14.02(02)	-0.64(01)	-0.66(01)	-0.65(03)	0.48(02)	-14.29(04)	-14.21
S30	24.09(00)	-24.66(02)	-0.80(01)	0.93(03)	1.07(05)	-7.70(04)	-7.06(06)	-8.45

¹C. Drischler, K. Hebeler, and A. Schwenk, Phys. Rev. Lett. 122, 042501 (2019)

DHS interactions using MBPT(4)



Summary

- The paper develops an automated, GPU-accelerated high-order MBPT framework for the nuclear EOS, reaching fifth order at the normal-ordered two-body level and including residual 3N contributions explicitly up to third order.
- For the soft Hebeler interactions, MBPT converges rapidly in PNM and remains overall controlled in SNM; fifth-order corrections are typically very small.
- These fifth-order corrections do not resolve the discrepancy between the predicted and empirical saturation point, suggesting that the limitation mainly comes from the interactions themselves, not simply from missing many-body orders.
- For the harder DHS interactions, the expansion is still reasonably controlled at low to moderate densities, but high-density SNM shows signs of a possible MBPT breakdown and needs nonperturbative benchmarks.